

TRENDALENS 2026

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Bridging the Language Gap

Building an Africa Language Machine Translation:
English–Dholuo translation

PROJECT TYPE

Capstone Presentation

FOCUS AREA

NLP & Social Impact

PROBLEM

THE LANGUAGE BARRIER IN DIGITAL HEALTHCARE

5,000,000+

Dholuo Speakers

A vast population in Kenya primarily uses Dholuo for daily communication, yet remains digitally underserved.

0% SUPPORT

Current Platforms

Major Kenyan telemedicine providers like BYON8 lack native support for Dholuo consultations.

Health Inequity

Language barriers lead to misdiagnosis and poor adherence to treatment among rural and elderly patients.

Trust Deficit

Patients are less likely to adopt digital health solutions if they cannot communicate in their primary tongue.

The Challenge: Bridging the gap between modern medical technology and the linguistic reality of East Africa.

OBJECTIVE

TRENDLENS: SPECIALIZED MEDICAL TRANSLATION

Develop a robust, clinical-grade translation pipeline between English and Dholuo to empower healthcare providers and patients.

TECHNICAL GOAL

Domain Expertise

Outperform general-purpose translation models in the specific context of medical terminology and clinical phrasing.

BUSINESS GOAL

Plug-and-Play API

Provide a scalable integration for telemedicine providers to expand their reach to millions of underserved speakers.

SUCCESS METRICS AND TARGET AUDIENCE

SUCCESS METRICS

- BLEU (Bilingual Evaluation Understudy)- produces score between 0-1
Higher BLEU scores
- chrF (Character F-score)
Score range 0 to 1

TARGET AUDIENCE

- Dholuo speaking patients
- Healthcare providers
- Kenyan telemedicine platform and providers

DATA: THE FOUNDATION OF LOW-RESOURCE NLP

DATASET SIZE

136,625

Parallel sentence pairs in English and Dholuo, providing a robust base for supervised learning.

DATA SOURCES

Hybrid

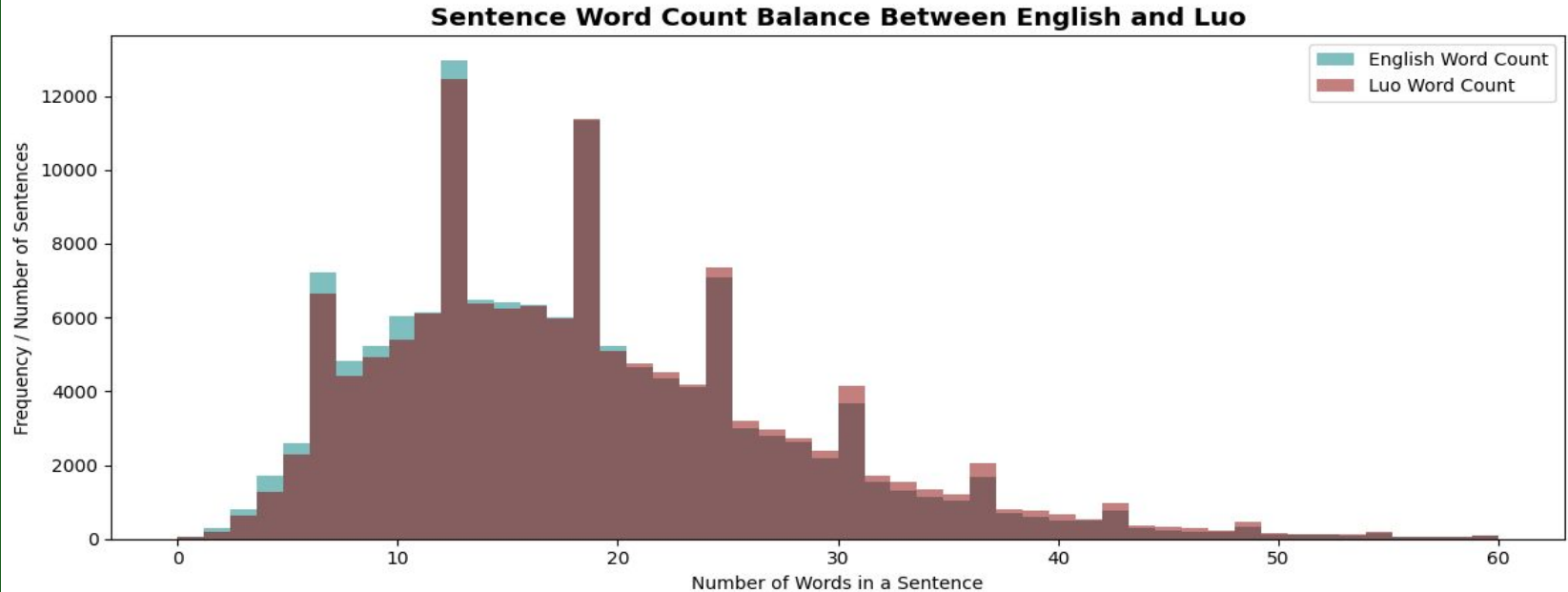
Combination of general-domain corpora and curated clinical phrases for domain specialization.

Processing Strategy

Rigorous cleaning to remove noise, followed by subword tokenization to handle Dholuo's complex morphology.

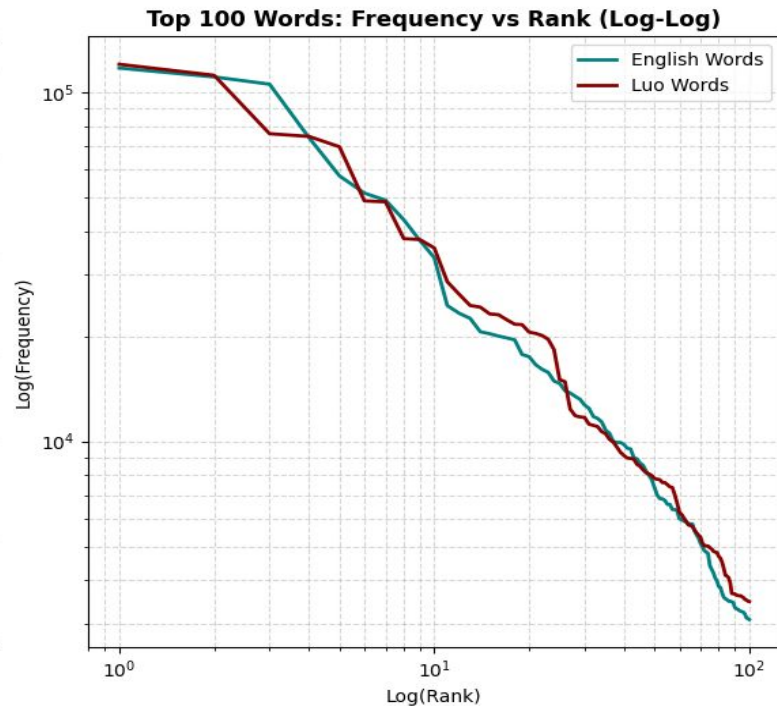
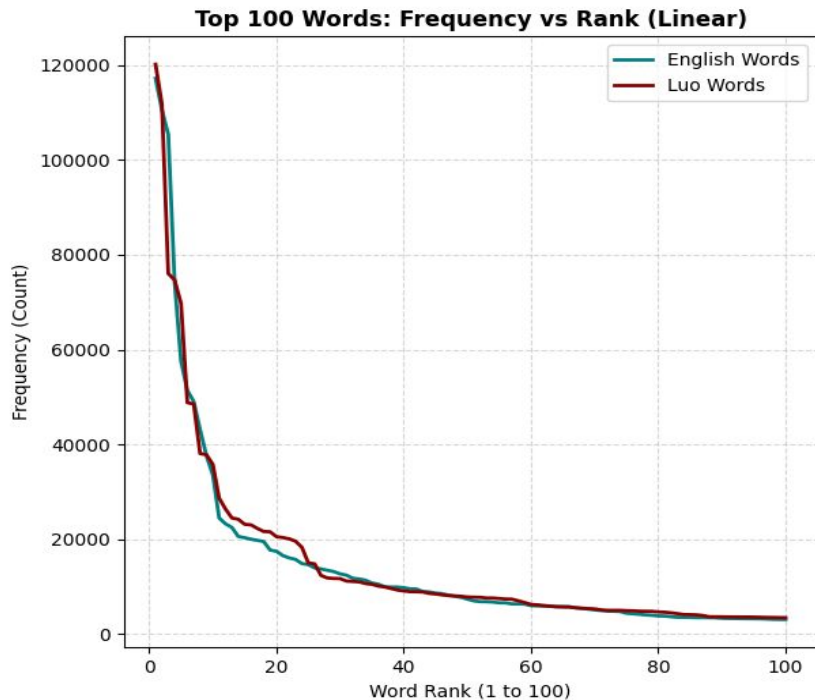
CLEANING > TOKENIZATION > ALIGNMENT > AUGMENTATION

SENTENCE WORD COUNT BALANCE BETWEEN ENGLISH AND LUO



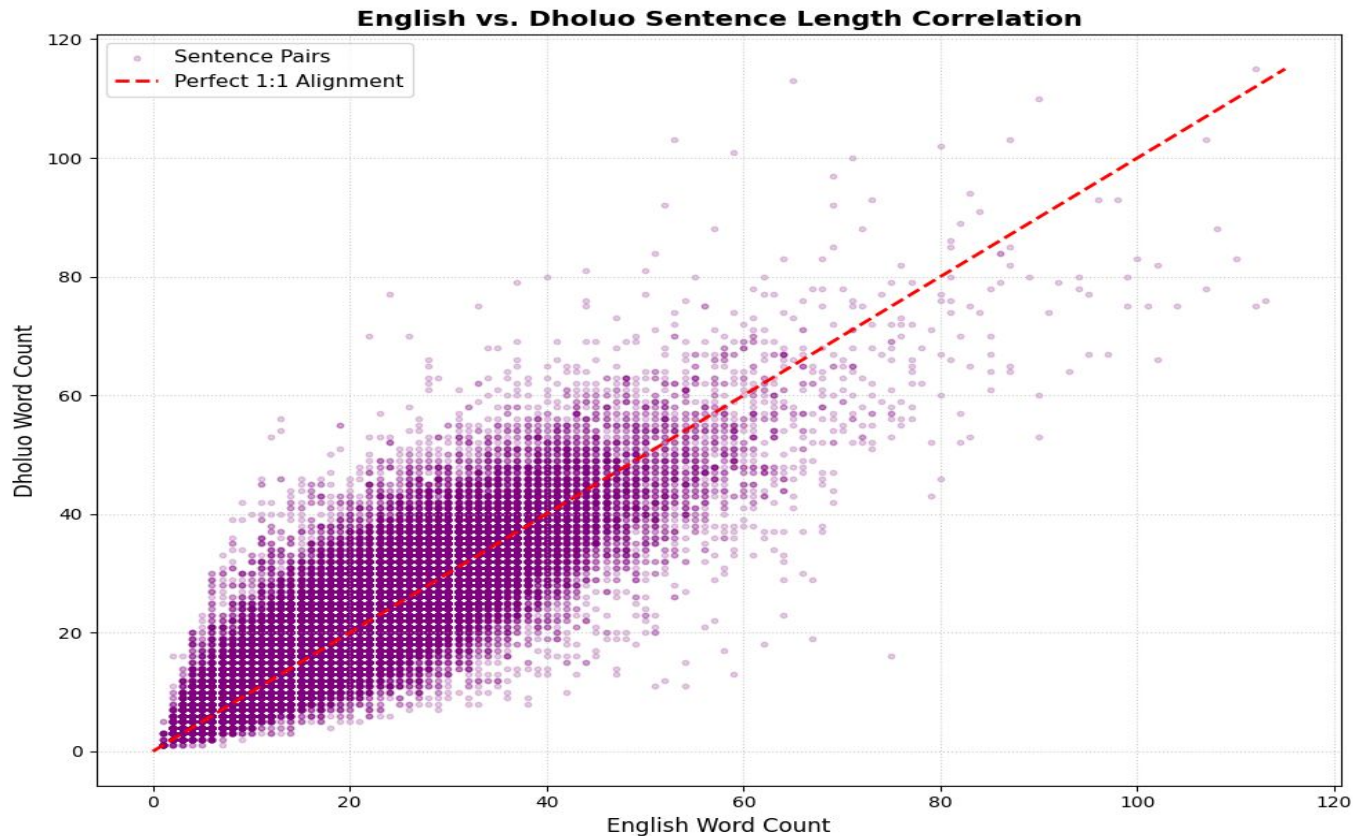
1. Luo words have an expansion Factor/ratio of 1.08
2. English sentences average 18.3 words , while Luo sentences average 19.1 words.

Zipf Plot



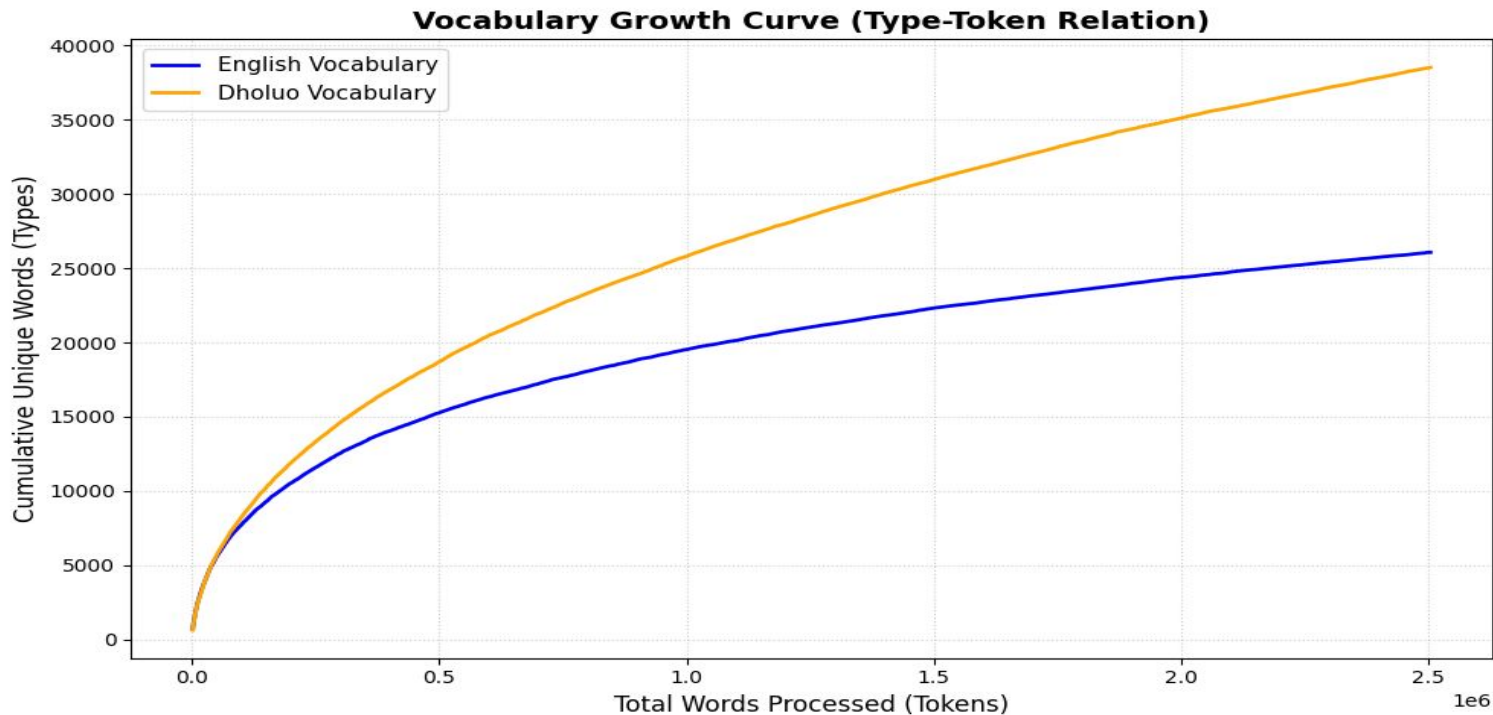
1. The tall bars on the left graph shows tokens words like "*the, to, of, and;* in" Dholuo *e, ne, ma, ni, gi*.
2. Dholuo is more morphologically rich and encodes grammatical information inside words rather than through a small set of standalone function words.

Sentence Length Correlation



1. Outliers to flag for cleaning, there are outlier mismatch that need cleaning.
2. Most sentence length scale together at close to a 1:1 rate, showing high correlation.

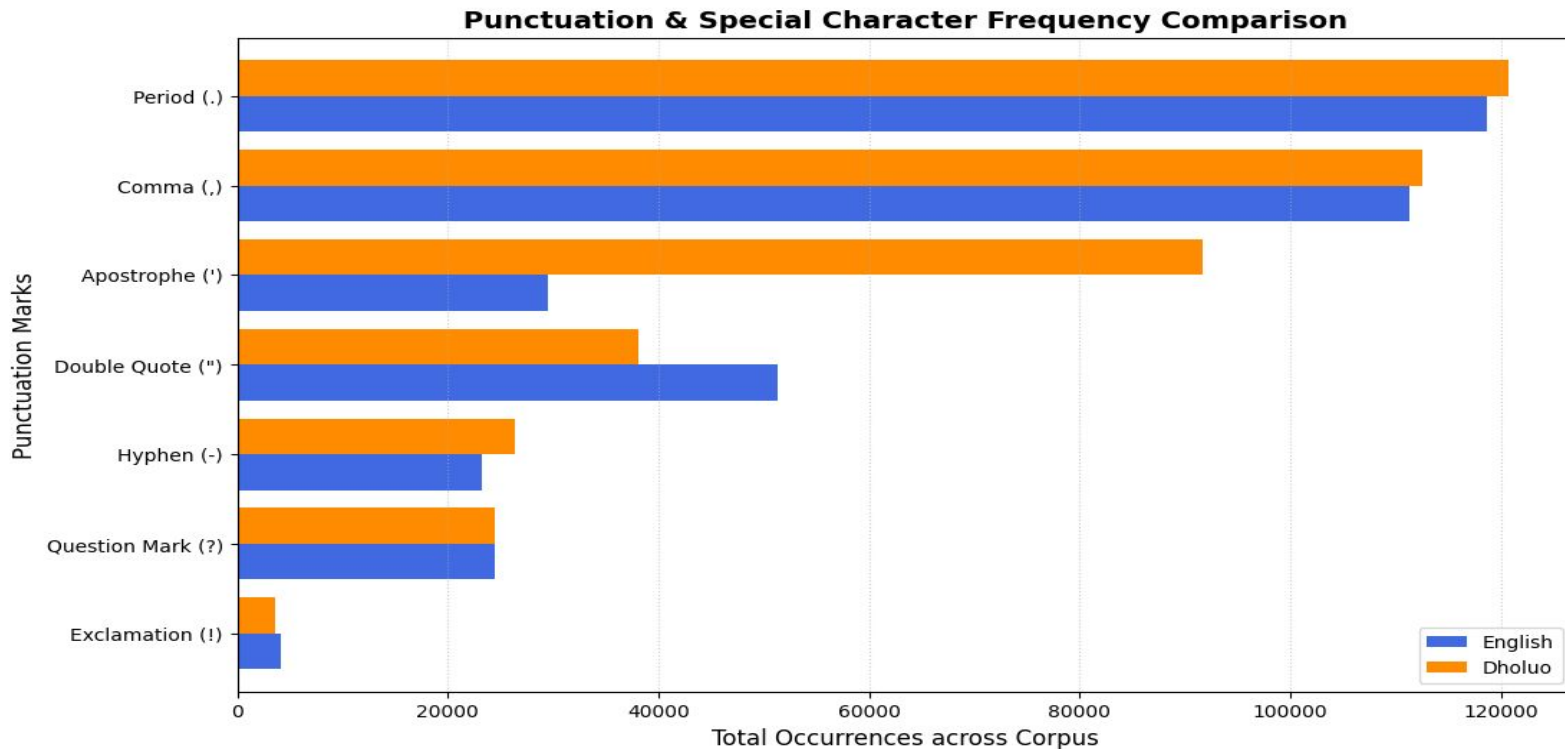
Vocabulary Growth Curve



1. Dholuo's line grows faster than English's this evidence of Dholuo's morphological richness.
2. Dholuo's type-token curve is steeper, thus the tokenizer needs a larger vocabulary.

EDA: Punctuation & Special Character

Punctuation & Special Character



1. **"The apostrophe anomaly"**, In Dholuo it is a functional letter and has consonant sounds (e.g. *ng'ato*, *gikone*, *wang'e*), stripping it changes the meaning of the word.
2. **Strong sentence-level alignment**: periods and commas occur at almost matching rate.

THE TECHNICAL PIPELINE: BASELINES TO SOTA

STAGE 01: BASELINES

TF-IDF + 1-NN & Seq2Seq LSTM

Initial benchmarks using statistical and RNN approaches.

STAGE 02: INTERMEDIATE

Seq2Seq + Attention & Transformer

Assessing attention and self-attention impact.

STAGE 03: STATE-OF-THE-ART

NLLB + LoRA (Low-Rank Adaptation)

Meta's multilingual model with efficient fine-tuning.

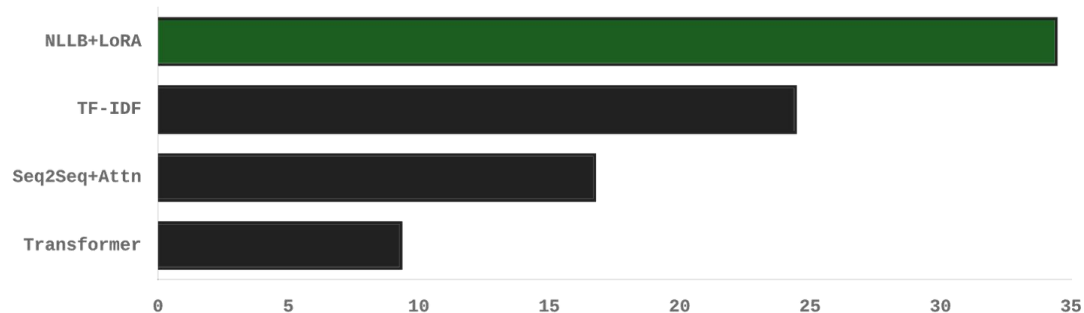
COMPUTE ENVIRONMENT:

High-performance training on Kaggle T4 x2 GPUs.

COMPARATIVE PERFORMANCE ANALYSIS

MODEL ARCHITECTURE	BLEU	CHRF	TRAIN TIME
NLLB + LoRA	34.49	50.76	62.8m
Seq2Seq + Attention	16.80	33.40	133.5m
TF-IDF + 1-NN	24.50	N/A	< 1m
Transformer (Baseline)	9.38	22.54	185.0m

VISUALIZING THE PERFORMANCE GAP (BLEU SCORE)



RESULTS

PERFORMANCE BREAKTHROUGH: NLLB + LORA

BLEU SCORE

34.49

Vastly outperforming the baseline
Transformer (9.38) and Seq2Seq models.

CHRF SCORE

50.76

High character-level overlap indicating superior
syntactic fluency in Dholuo.

TRAINING TIME

62.8M

3x faster than standard Transformer training
using Parameter-Efficient Fine-Tuning.

Training loss

1.168

Exceptionally low loss demonstrating
robust convergence and model stability.

BEYOND HEALTHCARE: A GENERAL FRAMEWORK



SECTOR 01: FINTECH

Financial Inclusion

Localizing mobile banking and micro-loan apps to drive financial literacy and trust in rural markets.



SECTOR 02: E-COMMERCE

Digital Trade

Enabling regional trade by translating product listings and customer support into local dialects.



SECTOR 03: PUBLIC SERVICE

Access to Justice

Rapid translation of government advisories and legal documents to ensure citizen rights and safety.

TrendLens is a foundational infrastructure for inclusive digital growth across East Africa.

THE BUSINESS VALUE OF LINGUISTIC INCLUSIVITY

MARKET EXPANSION

The Next Billion

Accessing underserved regional markets requires speaking the user's primary language, not just offering a service.

CUSTOMER LOYALTY

Trust & Retention

Users are significantly more likely to trust and remain loyal to platforms that communicate in their native tongue.

ESG IMPACT

Digital Equity

Fulfilling corporate social responsibility by bridging the digital and linguistic divide in developing economies.

RETURN ON INVESTMENT

Efficiency Gains

Low-cost, high-speed fine-tuning (LoRA) makes high-quality translation affordable for startups and SMEs.

3X FASTER ROI

ROADMAP

FUTURE ROADMAP: SCALING THE IMPACT

PHASE 01

Language Expansion

Reproducing the NLLB+LoRA pipeline for other Kenyan languages including Luhya, Kikuyu, and Kalenjin.

PHASE 02

Voice Integration

Developing speech-to-speech translation capabilities for real-time oral medical consultations.

PHASE 03

Edge Deployment

Optimizing models for mobile devices to ensure functionality in low-connectivity rural areas.

PHASE 04

Collaborations

Partnering with Health Ministries and NGOs to integrate TrendLens into public health infrastructure.



TECHNOLOGY FOR HUMAN CONNECTION

TECHNICAL VICTORY

NLLB + LoRA Success

Proven success of parameter-efficient fine-tuning for high-quality low-resource translation.

BUSINESS VICTORY

Scalable Inclusion

A foundational framework for digital equity across healthcare, finance, and trade.

Language should be a bridge, not a barrier, to human progress and global health equity.